

NutriLect AI Dietologist

Artificial Intelligence in Support of Child Nutrition: From Scientific Evidence to Personalized Recommendations

diet-autism-kids.bg

ICT CLUSTER VARNA

BSMEPA
BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

BRAIN++
AI FACTORY

ARC FUND

CONNEXUS

From one-size-fits-all diets to intelligent nutrition support for vulnerable children.

We transform pediatric nutrition from fragmented advice into data-driven, adaptive, personalized care.

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Why We Started

- ✓ Built from real clinical and research experience
- ✓ Inspired by the daily challenges of children with complex nutritional needs — autism, cerebral palsy, selective eating
- ✓ Designed to bridge the gap between data, decision-making and family support

This idea did not begin with technology. It began with practice.



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Problem Statement

Children with complex nutritional needs are often managed through:

- ✓ **Fragmented data** — dietary intake, growth, symptoms and labs sit in separate places
- ✓ **Delayed nutrition adjustments** — plans change late, after problems appear
- ✓ **Limited personalization** — static, one-size-fits-all advice
- ✓ **High burden on families and clinicians** — time, stress and uncertainty

Current reality: nutrition decisions are reactive rather than proactive.

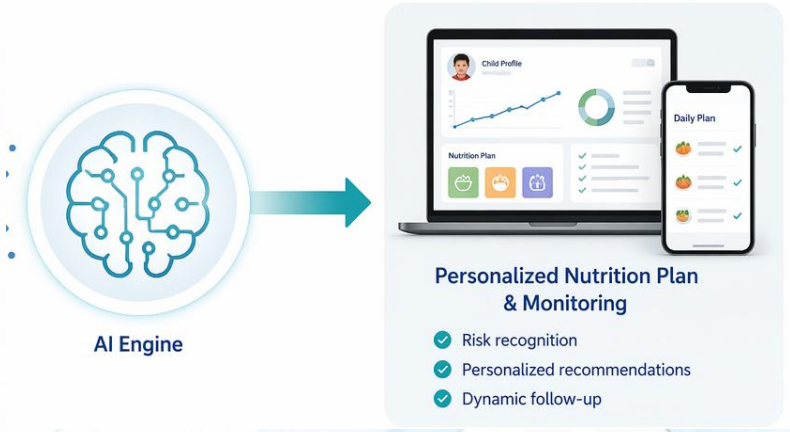


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Our Solution

Clinician-guided AI nutrition decision support

- ✓ Integrates clinical and nutrition-related data
- ✓ Supports early recognition of nutritional risk
- ✓ Generates personalized recommendations
- ✓ Enables dynamic follow-up over time



1. Assess
Collect & evaluate data

2. Stratify
Identify needs & risk

3. Personalize
Generate tailored recommendations

4. Monitor
Track progress in real time

5. Adapt
Adjust plan as needs change

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Our Platform

Live now at diet-autism-kids.bg — science-based nutrition guidance for children, in Bulgarian

1 AI Assistant

Interactive chat for quick answers and recommendations, built on NutriLect expertise and carefully selected scientific sources.

2 Detailed Questionnaire

Takes a little longer, but gives a detailed analysis and more personalized recommendations for the child's needs.

3 Useful Information

Plain-language evidence summaries of elimination diets (gluten-free, casein-free, ketogenic, SCD, GAPS) and an FAQ.

For every child — with special support for children with autism, on elimination diets, or with very restrictive eating.

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Our Team



Pediatrics · Pediatric nutrition · Translational research · Digital health · Biomarkers & microbiome

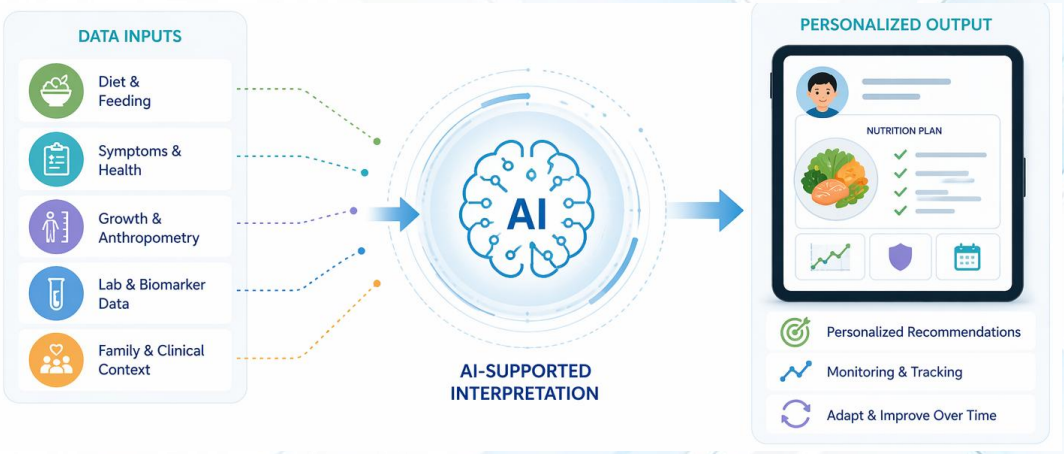
Funded by the European Union
NextGenerationEU

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How It Works

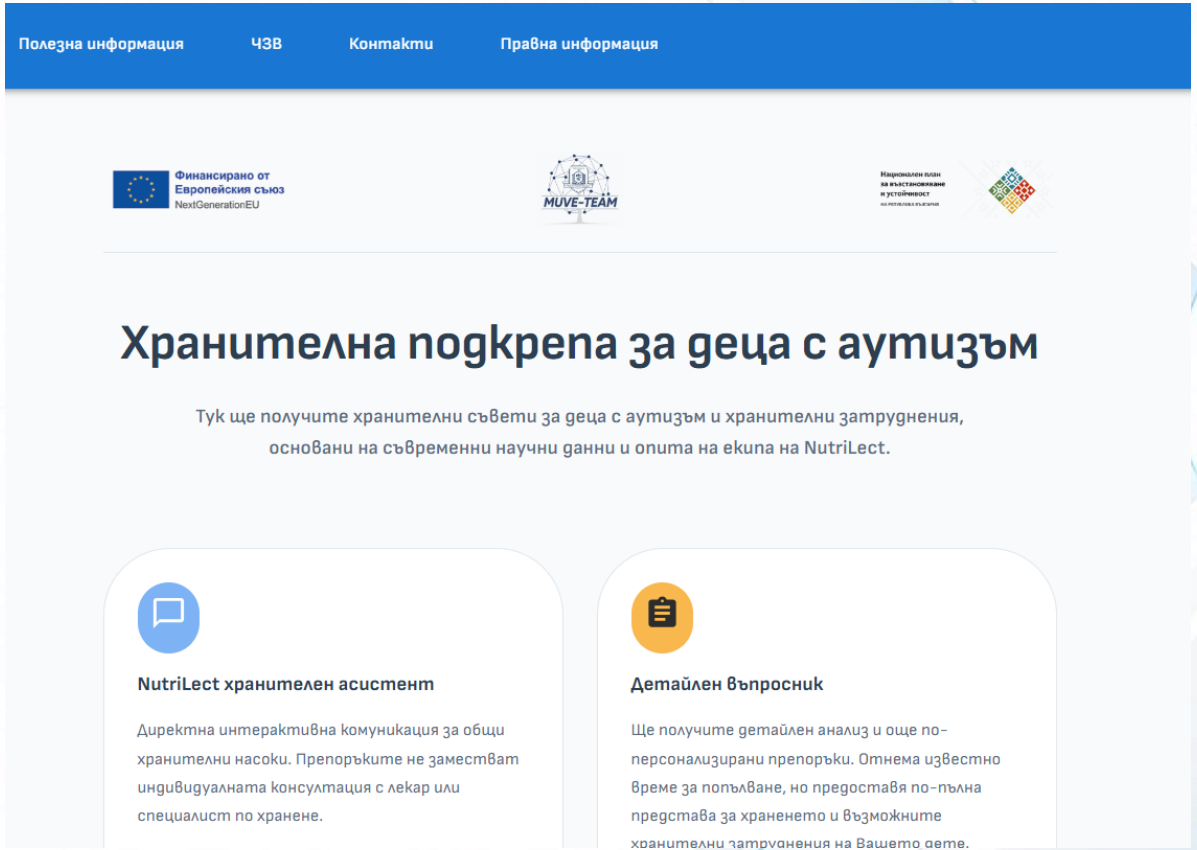
Data inputs

- ✓ Dietary intake
- ✓ Symptoms & feeding difficulties
- ✓ Growth & anthropometry
- ✓ Clinical context
- ✓ Family preferences



Data → AI-supported interpretation → personalized recommendation & monitoring

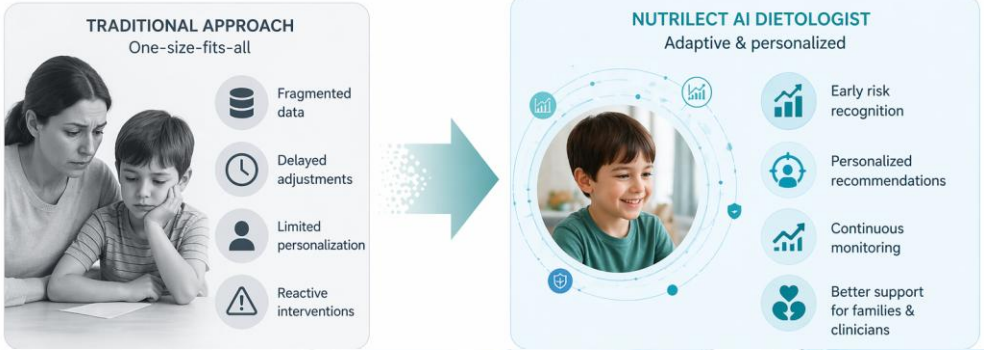
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Полезна информация
ЧЗВ
Контакти
Правна информация
Финансирано от Европейския съюз
NextGenerationEU
MUVE-TEAM
Национален план за възстановяване и устойчивост
Хранителна подкрепа за деца с аутизъм
Тук ще получите хранителни съвети за деца с аутизъм и хранителни затруднения, основани на съвременни научни данни и опита на екипа на NutriLect.
NutriLect хранителен асистент
Директна интерактивна комуникация за общи хранителни насоки. Препоръките не заместват индивидуалната консултация с лекар или специалист по хранене.
Детайлен въпросник
Ще получите детайлен анализ и още по-персонализирани препоръки. Отнема известно време за попълване, но предоставя по-пълна представа за храненето и възможните

Why It Matters

From reactive to proactive care

- ✓ Faster recognition of nutritional risk
- ✓ More personalized recommendations
- ✓ Better longitudinal monitoring
- ✓ Stronger bridge between research and real-world care



Future extension

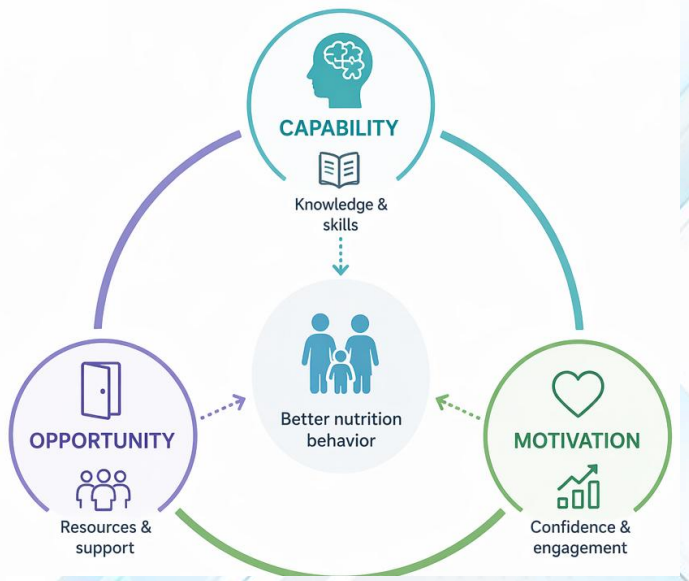
Microbiome-guided nutritional innovation, such as targeted prebiotic concepts for children with autism, informed by our ongoing microbiota analyses.

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Behavioral Value: COM-B

Helping families change everyday eating behavior

- **C** — **Capability** — personalized, understandable guidance
- **O** — **Opportunity** — structured support and monitoring
- **M** — **Motivation** — visible progress and adaptive feedback



Model: Michie, van Stralen & West, UCL Centre for Behaviour Change

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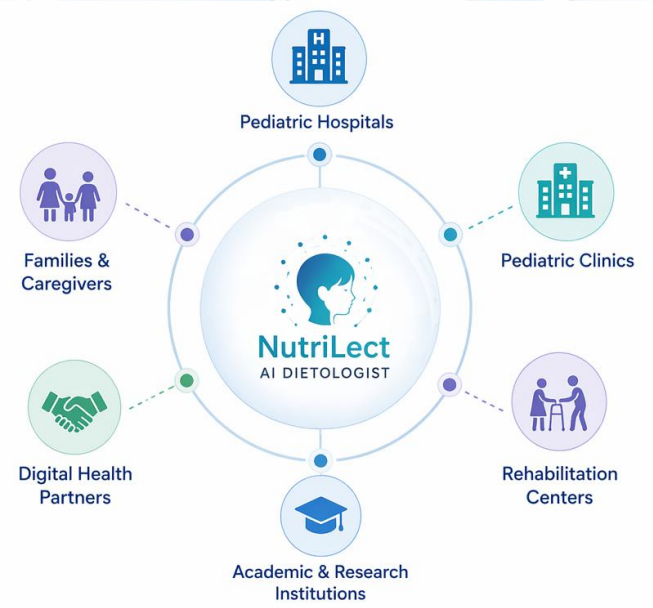
Go to Market

Target customers

- ✓ Pediatric clinics and hospitals
- ✓ Child development & rehabilitation centers
- ✓ Academic health centers
- ✓ Digital health partners

Business logic

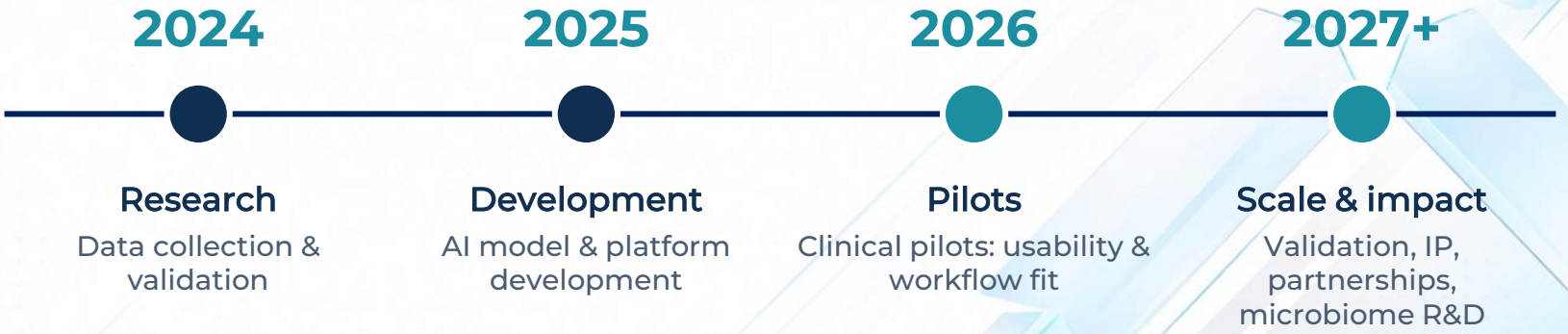
- ✓ Pilot partnerships
- ✓ Institutional licensing
- ✓ Subscription-based access
- ✓ Modular expansion



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Roadmap

Next 18 months: refine prototype → clinical pilots → validation & partnerships





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